

Module 1

Kotlin Deep Dive

Type System & Language Core

- Declaration-site variance & Star projection
- Platform Types & Intersection Types
- Sealed Classes/Interfaces & Exhaustive when
- Contracts
- Operator Overloading & Convention Methods
- Destructuring Internals
- Infix Functions & Extension Function Resolution

Functions & Functional Style

- inline / noinline / crossinline — Bytecode Impact
- Scope Functions — When Each Actually Fits
- Local Functions & Closures & tailrec

Classes, Initialization & Lifecycle

- Initialization Order
- lateinit Internals
- Delegated Properties
- companion object vs Top-Level & const

Equality, Immutability & State

- `==` vs `===` & `equals/hashCode` Contracts
- `val` $\neq$ `Immutable`
- The Cast Attack: `List` $\rightarrow$ `MutableList`
- Backing Properties Beyond Flow

Collections & Performance

- Sequences vs Collections
- Primitive Arrays
- ArrayDeque & Persistent Collections
- Lesser-Known Operators

Coroutines

- Suspend Functions & Builders
- Structured Concurrency & Job Hierarchy
- Dispatchers & Context Switching
- Cooperative Cancellation
- Timeouts & Retry
- Exception Handling

Coroutines: Flow & Beyond

- Flow Basics: Builders & Operators
- Flow: Cold/Hot & Backpressure
- Bridging Callback APIs
- Mutex & Semaphore vs synchronized
- Coroutine Internals: CPS & State Machines

Error Modeling

- `Result<T>` & the `runCatching` Trap
- Checked Exceptions Interop & `@Throws`
- Typed Errors: Either-Style

Modern Kotlin & Ecosystem

- K2 Compiler & Smarter Smart Casts
- New Language Features (2.2+)
- Multiplatform Awareness